9.1 The Metric System

Figure 9.2 A scale that measures weight in both metric and customary units. (credit: “Weighing the homemade cheddar” by Ruth Hartnup/Flickr, CC BY 2.0)

Learning Objectives

After completing this section, you should be able to:

1. Identify units of measurement in the metric system and their uses.
2. Order the six common prefixes of the metric system.
3. Convert between like unit values.

Even if you don’t travel outside of the United States, many specialty grocery stores utilize the metric system. For example, if you want to make authentic tamales you might visit the nearest Hispanic grocery store. While shopping, you discover that there are two brands of masa for the same price, but one bag is marked 1,200 g and the other 1 kg. Which one is the better deal? Understanding the metric system allows you to understand that 1,200 grams is equivalent to 1.2 kg, so the 1,200 g bag is the better deal.

Units of Measurement in the Metric System

Units of measurement provide common standards so that regardless of where or when an object or substance is measured, the results are consistent. When measuring distance, the units of measure might be feet, **meters**, or miles. Weight might be expressed in terms of pounds or **grams**. Volume or capacity might be measured in gallons, or **liters**. Understanding how metric units of measure relate to each other is essential to understanding the metric system:

- The metric unit for distance is the meter (m). A person’s height might be written as 1.8 meters (1.8 m). A meter slightly longer than a yard (3 feet), while a centimeter is slightly less than half an inch.
- The metric unit for area is the **square meter** (m^2). The area of a professional soccer field is 7,140 square meters (7,140 m^2).
- The metric base unit for volume is the **cubic meter** (m^3). However, the liter (L), which is a metric unit of capacity, is used to describe the volume of liquids. Soda is often sold in 2-liter (2 L) bottles.
- The gram (g) is a metric unit of mass but is commonly used to express weight. The weight of a paper clip is approximately 1 gram (1 g).
- The metric unit for temperature is **degrees Celsius** (°C). The temperature on a warm summer day might be 24 °C.

While the U.S. Customary System of Measurement uses ounces and pounds to distinguish between weight units of different sizes, in the metric system a *base unit* is combined with a prefix, such as *kilo–* in kilogram, to identify the relationship between smaller or larger units.

When using abbreviations to represent metric measures, always separate the quantity and the units with a space, with no spaces between the letters or symbols in the units. For example, 7 millimeters is written as 7 mm, not 7mm.

TECH CHECK

It is important to be able to convert between the U.S. Customar System of Measurement and the metric system. However, in this chapter we'll focus on converting units within the metric system. Why? Typing "200 centimeters in inches" into any browser search bar will instantly convert those measures for you (Figure 9.3). You'll have an opportunity in the Projects to work between measurement systems.

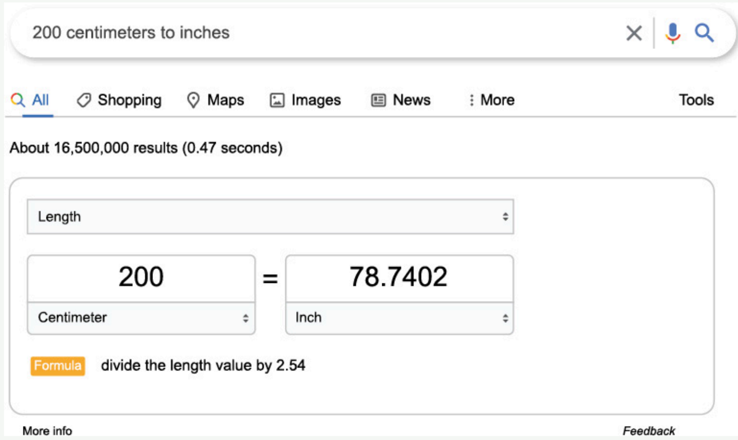

Figure 9.3 (credit: Screenshot/Google)

Let's be honest. Most of us use computers or smartphones to perform many of the calculations and conversions we were taught in math class. But there is value in understanding the metric system since it exists all around us, and most importantly, knowing how the different metric units relate to each other allows you to compare prices, find the right tool in a workshop, or acclimate when in another country. No matter the circumstance, you cannot avoid the metric system.

EXAMPLE 9.1

Determining the Correct Base Unit

Which base unit would be used to express the following?

1. amount of water in a swimming pool
2. length of an electrical wire
3. weight of one serving of peanuts

Solution

1. Liquid volume is generally expressed in units of liters (L).
2. Length is measured in units of meters (m).
3. Weight is commonly expressed in units of grams (g).

YOUR TURN 9.1

Determine the correct base measurement for each of the following.

1. weight of a laptop
2. width of a table
3. amount of soda in a pitcher

While there are other base units in the metric system, our discussions in this chapter will be limited to units used to express length, area, volume, weight, and temperature.

Metric Prefixes

Unlike the U.S. Customary System of Measurement in which 12 inches is equal to 1 foot and 3 feet are equal to 1 yard, the metric system is structured so that the units within the system get larger or smaller by a power of 10. For example, a centimeter is 10^{-2}, or 100 times smaller than a meter, while the kilometer is 10^3, or 1,000 times larger than a meter.

The metric system combines base units and unit prefixes reasonable to the size of a measured object or substance. The most used prefixes are shown in Table 9.1. An easy way to remember the order of the prefixes, from largest to smallest, is the mnemonic *King Henry Died From Drinking Chocolate Milk.*

Prefix	kilo–	hecto–	deca–	base unit	deci–	cent–	milli–
Abbreviation	k	h	da		d	c	m
Magnitude	10^3	10^2	10^1	10^0 or 1	10^{-1}	10^{-2}	10^{-3}

Table 9.1 Metric Prefixes

EXAMPLE 9.2

Ordering the Magnitude of Units
Order the measures from smallest unit to largest unit.

centimeter, millimeter, decimeter

Solution
Looking at the metric prefixes, we can see that the prefix order from smallest unit to largest unit is milli-, centi-, deci-, so the order of the units from smallest to largest is millimeter, centimeter, decimeter.

YOUR TURN 9.2

1. Order the measures from largest unit to smallest unit.
   hectogram, decagram, kilogram

EXAMPLE 9.3

Determining Reasonable Values for Length
What is a reasonable value for the length of a person's thumb: 5 meters, 5 centimeters, or 5 millimeters?

Solution
Given that a meter is slightly longer than a yard, 5 meters is not a reasonable value for the length of a person's thumb. Since a millimeter is 10 times smaller than a centimeter, which is approximately $\frac{1}{2}$ inch, 5 millimeters is not a reasonable estimate for the length of a person's thumb. The correct answer is 5 centimeters.

YOUR TURN 9.3

1. What is a reasonable estimate for the length of a hallway: 2.5 kilometers, 2.5 meters, or 2.5 centimeters?

Converting Metric Units of Measure

Imagine you order a textbook online and the shipping detail indicates the weight of the book is 1 kg. By attaching the letter "k" to the base unit of gram (g), the unit used to express the measure is 10^3 or 1,000 times greater than a gram. One kilogram is equivalent to 1,000 grams.

The tip of a highlighter measures approximately 1 cm. The letter "c" attached to the base unit of meter (m) means the unit used to express the measure is 10^{-2} or $\frac{1}{100}$ of a meter. One meter is equivalent to 100 centimeters.

A **conversion factor** is used to convert from smaller metric units to bigger metric units and vice versa. It is a number that when used with multiplication or division converts from one metric unit to another, both having the same base unit. In the metric system, these conversion factors are directly related to the powers of 10. The most common used conversion factors are shown in Figure 9.4.

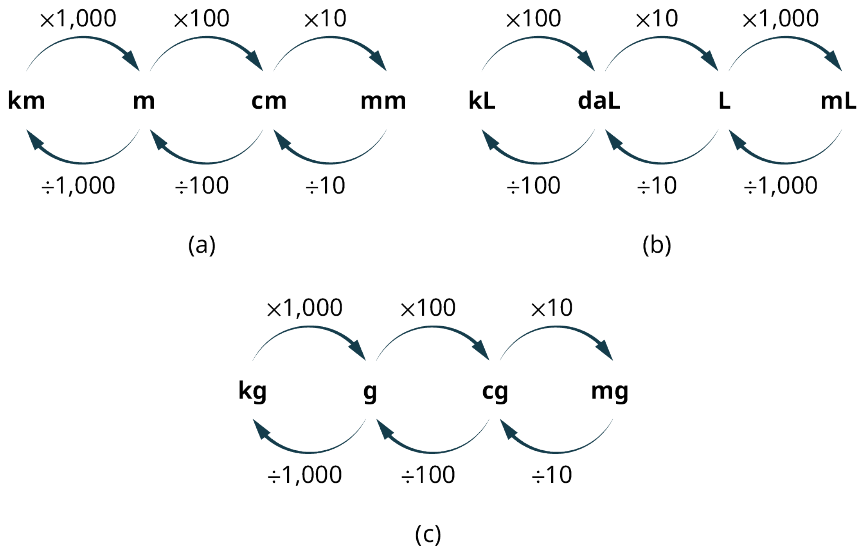

Figure 9.4 Common Metric Conversion Factors for (a) Meters, (b) Liters, and (c) Grams

EXAMPLE 9.4

Converting Metric Distances Using Multiplication

The firehouse is 13.45 km from the library. How many meters is it from the firehouse to the library?

Solution

When converting from a larger unit to a smaller unit, use multiplication. The conversion factor from kilometer to the base unit of meters is 1,000.

$$13.45 \times 1{,}000 = 13{,}450$$

So, the firehouse is 13,450 meters away from the firehouse.

YOUR TURN 9.4

1. The record for the men’s high jump is 2.45 m. What is the record when expressed in centimeters?

EXAMPLE 9.5

Converting Metric Capacity Using Division

How many liters is 3,565 milliliters?

Solution

When converting from a smaller unit to a larger unit, use division. The conversion factor from milliliter to the base unit of liters is 1,000.

$$3{,}565 \div 1{,}000 = 3.565$$

So, 3,565 milliliters is 3.565 liters.

YOUR TURN 9.5

1. A bottle of cleaning solution measures 7.6 liters. How many decaliters is that?

EXAMPLE 9.6

Converting Metric Units of Mass to Solve Problems

Caroline and Aiyana are working on a chemistry experiment together and must perform calculations using measurements taken during the experiment. Due to miscommunication, Caroline took measurements in centigrams and Aiyana used milligrams. Convert Caroline's measurement of 125 centigrams to milligrams.

Solution

When converting from a larger unit to a smaller unit, use multiplication. The conversion factor from centigrams to the milligrams is 10.

$$125 \times 10 = 1{,}250$$

So, the 125 centigrams are 1,250 milligrams.

YOUR TURN 9.6

1. Convert Aiyana's measurement of 1,457 mg to centigrams.

EXAMPLE 9.7

Converting Metric Units of Volume to Solve Problems

A bottle contains 500 mL of juice. If the juice is packaged in 24-bottle cases, how many liters of juice does the case contain?

Solution

Step 1: Multiply the amount of juice in each bottle by the number of bottles.

$$500 \text{ mL} \times 24 = 12{,}000 \text{ mL}$$

Step 2: Divide by 1,000 to convert from milliliters to liters.

$$\frac{12{,}000 \text{ mL}}{1{,}000} = 12 \text{ L}$$

So, there are 12 liters of juice in each case.

YOUR TURN 9.7

1. A hospital orders 250 doses of liquid amoxicillin. Each dose is 5 mL. How many liters of amoxicillin did the hospital order?

PEOPLE IN MATHEMATICS

Valerie Antoine

In the 1970s, people were told that they must learn the metric system because the United States was soon going to convert to using metric measurements. Children and young adults probably watched educational cartoons about the metric system on Saturday mornings.

In 1975, President Gerald Ford signed the Metric Conversion Act and created a board of 17 people commissioned to coordinate the voluntary switch to the metric system in the United States. Among those 17 people was Valerie Antoine, an engineer who made it her life's work to push for this change. Despite President Ronald Reagan dissolving the board in 1982, effectively killing the move to the metric system at the time, Antoine continued the movement out of her own home as the executive director of the U.S. Metric Association. Reagan's decision followed intensive lobbying by American businesses whose factories used machinery designed to use customary measurements by workers trained in customary measurements. There was also intense public pressure from American citizens who didn't want to go through the time consuming and expensive process of changing the country's entire infrastructure. Fueled by a Congressional mandate in 1992 that required all federal agencies make the switch to the metric system, Antoine never gave up hope that the metric system would trickle down from the government and find its way into American schools, homes, and everyday life.

VIDEO

U.S. Office of Education: Metric Education (https://openstax.org/r/U.S._Office_Education)

EXAMPLE 9.8

Converting Grams to Solve Problems

The nutrition label on a jar of spaghetti sauce indicates that one serving contains 410 mg of sodium. You have poured two servings over your favorite pasta before recalling your doctor's advice about keeping your sodium consumption below 1 g per meal. Have you followed your doctor's recommendation?

Solution

Step 1: Multiply the number of servings by the amount of sodium in each serving.

$$410 \text{ mg} \times 2 = 820 \text{ mg}$$

Step 2: Divide by 1,000 to convert from milligrams to grams.

$$\frac{820 \text{ mg}}{1{,}000} = 0.82 \text{ g}$$

You have followed doctor's recommendation because 0.82 g is less than 1 gram.

YOUR TURN 9.8

1. The FDA recommends that you consume less than 0.5 g of caffeine daily. A cup of coffee contains 95 mg of caffeine and a can of soda contains 54 mg. If you drink 2 cups of coffee and 3 cans of soda, have you kept your day's caffeine consumption to the FDA recommendation? Explain.

EXAMPLE 9.9

Comparing Different Units

A student carefully measured 0.52 cg of copper for a science experiment, but their lab partner said they need 6 mg of copper total. How many more centigrams of copper does the student need to add?

Solution

Step 1: Convert these two measurements into a common unit. Since the question asks for the number of centigrams, convert 6 mg to centigrams, which is 0.6 cg.

Step 2: Find the difference by subtracting $0.6 - 0.52$ which is 0.08 cg. This means the student must add another 0.08 cg of copper.

YOUR TURN 9.9

1. Kyrie boasted he jumped out of an airplane at an altitude of 3,810 meters on his latest skydive trip. His friend said they beat Kyrie because their jump was at an altitude of 3.2 km. Whose skydive was at a greater altitude?

WHO KNEW?

The United States and the Metric System

Did you know that the metric system pervades daily life in the United States already? While Americans still may purchase gallons of milk and measure house sizes in square feet, there are many instances of the metric system. Photographers buy 35 mm film and use 50 mm lenses. When you have a headache, you might take 600 mg of ibuprofen. And if you are eating a low-carb diet you probably restrict your carb intake to fewer than 20 g of carbs daily. Did you know even the dollar is metric? In the video, Neil DeGrasse Tyson and comedian co-host Chuck Nice provide an amusing perspective on the metric system.

The International System of Units (SI) is the current international standard metric system and is the most widely used system around the world. In most English-speaking countries SI units such as meter, liter, and metric ton are spelled metre, litre, and tonne.

VIDEO

Neil deGrasse Tyson Explains the Metric System (https://openstax.org/r/Tyson_Explains_Metric_System)

WORK IT OUT

Get to Know the Metric System
Just how much is the metric system a part of your life now? Probably more than you think. For the next 24 hours, take notice as you move through your daily activities. When you are shopping, are the package sizes provided in metric units? Change the weather app on your phone to display the temperature in degrees Celsius. Are you able to tell what kind of day it will be now? While the United States is not officially using the metric system, you will still find the metric system all around you.

Check Your Understanding

1. Which metric base unit would be used to measure the height of a door?
2. Which metric base unit would be used to measure your weight?
3. Which is greater: 12 hectoliters or 12 centiliters?
4. Convert 1,520 cm to meters (m).
5. Convert 1.34 km to decameters (dam).
6. Convert 12,700 cg to hectograms (hg).
7. Convert 750 km to millimeters (mm).
8. Which is the larger measurement: 0.04 dam or 40 cm?

SECTION 9.1 EXERCISES

For the following exercises, determine the base unit of the metric system described. Choose from *liter*, *gram*, or *meter*.

1. amount of soda in a bottle
2. weight of a book you are mailing
3. amount of gasoline needed to fill a car's tank
4. height of a computer desk
5. weight of a dog at the veterinarian's office
6. dimensions of the newest HD TV
7. distance a student athlete ran on the treadmill during their latest workout
8. amount of water to add to bleach for mopping floors

9. Write the order of the metric prefixes from greatest to least.

For the following exercises, choose the smaller of the two units.

10. decagrams or decigrams
11. centimeters or millimeters
12. liters or kiloliters
13. decigrams or centigrams
14. decameters or hectometers
15. milliliters or deciliters

16. Convert 158 hectometers to meters (m).

17. Convert 12.3 cg to grams (g).

18. Convert 160 dam to kilometer (km).

19. You purchase 10 kg of candy. You divide the candy into 25 bags. How many grams of candy are in each bag?

20. An outdoor track is 400 m long. If you run 10 laps around the track, how many kilometers have you run?

21. An aspirin tablet is 650 mg. If you take 2 aspirin twice in one day, how many grams of aspirin have you taken?

22. A juice box contains 450 mL of juice. If there are 6 juice boxes in a package, how many liters of juice are in the package?

23. Carlos consumed 4 cans of his favorite energy drink. If each can contains 111 mg of caffeine, how many grams of caffeine did he consume?

24. Celeste ran a total distance of 72,548 m in 1 week while training for a 5K fun run. How many kilometers did she run?

25. During week one of their diet, Dakota consumed 413 g of carbs. After speaking to their doctor, they only consumed 210 g of carbs the second week. How many fewer milligrams of carbs did they consume the second week?

26. Kaylea gained 2.3 kg of muscle weight in 9 months of working out. Cho gained 250 decagrams of muscle during that same time. Who gained more muscle weight? How many grams more?